MDCA-Net: Remote Sensing Image Classification Based on Multiscale Dynamic Convolution and Attention Enhancement

Xiangsuo Fan, Yan Zhang, Jinlong Fan, and Xuyang Li

***Abstract*—In response to the challenges faced by traditional deep learning (DL) models in adaptively extracting multiscale features from remote sensing images and establishing long-range dependencies across both channels and spatial dimensions—issues that contribute to a decline in classification performance—this study introduces a novel network architecture that integrates dynamic multiscale attention with spatiotemporal fusion, designated as multiscale dynamic convolution and attention fusion network (MDCA-Net). The research presents a dynamic multiscale large kernel attention, termed D-MLKA, which is engineered to dynamically weight and amalgamate multiscale convolutional features derived from 7×7 and 5×5 kernels, thereby facilitating a spectral–spatial adaptive perception mechanism. Subsequently, a spectral–spatial convolution and attention fusion module (SS-CAFM) is developed to leverage depthwise separable convolutions for the extraction of cross-channel spectral correlations, in conjunction with multihead attention to effectively capture long-range spatial dependencies. Moreover, a pseudo-temporal enhancement unit is introduced to bolster the model's robustness against interannual variations by generating virtual temporal features. Comparative experiments conducted on the Indian Pines and Houston University public datasets yielded overall accuracies (OAs) of 77.38% and 83.34%, respectively. Furthermore, when validated against the 2024 multispectral dataset from Rong'an County, Guangxi Autonomous Region, the model achieved a classification accuracy of 98.66%, representing a 2.15% improvement over existing methodologies such as Morphformer. Ultimately, the model demonstrated average accuracy (AA) of 98.12% and 97.79% on datasets from 2020 and 2022, respectively, thereby sustaining high performance levels while mitigating the risk of performance degradation.**

***Index Terms*—Attention enhancement, bidirectional long short-term memory (BiLSTM), multiscale dynamic convolution, remote sensing image classification.**

I. Introduction

HIGH-RESOLUTION remote sensing imagery has become indispensable in land-cover classification tasks that support environmental monitoring [1], disaster assessment [2], and intelligent urban development [3]. The growing availability of submeter spatial resolution and fused multispectral/hyperspectral observations from satellite and autonomous aerial vehicles (AAV) platforms [4] provides rich spatial textures and fine-grained spectral signatures, enabling more detailed modeling of complex land surface structures. However, the substantially increased dimensionality and variability of such data impose stringent requirements on the adaptability, robustness, and generalization capability of feature extraction models.

Unlike natural images, remote sensing imagery exhibits extreme multiscale heterogeneity. A single land-cover type may appear in dramatically different spatial extents—from pixel-level small objects such as shacks or narrow roads to large industrial complexes spanning tens of thousands of pixels. Conventional neural architectures designed for natural images, which rely on fixed receptive fields, are therefore prone to missing small targets or oversmoothing large structures [5]. Simultaneously, the nonlinear relationships between multispectral bands remain insufficiently explored; many existing methods rely on simple band stacking or dimensionality reduction, which may weaken the discriminative power of individual bands [6], [7]. Furthermore, modeling from a static single-time-phase perspective limits the ability to capture contextual dependencies in geographically complex areas [8]. These challenges emphasize the need for a model that can effectively integrate multiscale structural cues, spectral–spatial interactions, and contextual dependencies.

Deep learning (DL) techniques have achieved significant progress in remote sensing classification due to their ability to automatically learn hierarchical features from large datasets, reducing reliance on handcrafted feature engineering [9]. Although recurrent neural networks (RNNs) are effective for temporal sequence modeling [10], [11] [12], their sequential processing mechanism is not well-suited for capturing spatially structured information in 2-D remote sensing data [13]. This limitation motivates the shift toward models capable of handling both spatial heterogeneity and high-dimensional spectral information.

Among DL architectures, Transformer models have demonstrated strong global modeling capability through their

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self-attention mechanisms [14]. Vision transformer (ViT) [15] and subsequent variants such as SF-Net [16] have shown potential in hyperspectral image (HSI) classification. However, Transformers alone cannot fully address multiscale and local texture modeling, because self-attention tends to emphasize global relationships while being less effective in capturing fine spatial structures such as edges and boundaries. Conversely, CNNs possess strong locality modeling ability but are inherently constrained by limited receptive fields, making them insufficient for modeling long-range spatial dependencies inherent in remote sensing images.

These complementary characteristics have led to the development of hybrid CNN–Transformer architectures. Representative works include Morphformer [17], GCFormer [18], SSFTT [19], and CACFT [20], which integrate convolutional inductive biases with global attention to enhance spectral–spatial representation. However, applying Transformers to high-resolution images requires partitioning them into a large number of patches, resulting in excessively long sequences that significantly increase computational cost and memory demand [21]. Additionally, standard self-attention mechanisms still struggle to model local structural nuances compared with convolution-based context modeling.

In recent years, sequence modeling techniques have also attracted increasing attention in remote sensing applications, particularly for capturing contextual dependencies embedded in high-dimensional spectral features. Bidirectional long short-term memory (Bi-LSTM) networks, as an extension of RNNs, are capable of modeling contextual information in both forward and backward directions, thereby enhancing the representation of complex sequential dependencies [22]. Previous studies have demonstrated that such bidirectional structures can effectively capture intrinsic correlations in remote sensing data and improve feature discriminability by jointly considering past and future contextual cues [23]. However, directly applying temporal sequence models to spatially structured remote sensing images remains challenging, especially under single-temporal acquisition settings. This limitation motivates the exploration of pseudo-temporal modeling strategies that reinterpret spectral features as ordered sequences, enabling sequence-based models to be effectively integrated into spatial feature learning frameworks.

Motivated by these challenges, this study proposes the multiscale dynamic convolution and attention fusion network (MDCA-Net), designed to adaptively capture multiscale spatial structures, exploit cross-channel spectral dependencies, and enhance contextual modeling. The network first reorganizes the input multispectral data into 2-D spatial feature maps. These maps are then processed by two parallel modules: 1) a dynamic multiscale large kernel attention (D-MLKA) module, which adaptively fuses multiscale features using depthwise separable convolution with 7×7 and 5×5 kernels; and 2) a spectral–spatial convolutional and attention fusion module (SS-CAFM), which captures nonlinear spectral interactions through 3-D convolution and models long-range spatial dependencies using multihead attention. Their outputs are fused through residual aggregation. To further enhance the model's robustness to complex spatial patterns and subtle land-cover variations, we introduce a pseudo-temporal augmentation mechanism, which constructs virtual 3-D temporal sequences and employs bidirectional LSTMs and multihead attention to extract salient contextual features. Finally, a lightweight multilayer perceptron (MLP) classifier produces pixel-level predictions. This integrated design enables MDCA-Net to effectively address the limitations of existing approaches by providing adaptive multiscale representation, enriched spectral–spatial interaction modeling, and enhanced contextual awareness for fine-grained land-cover classification.

The main contributions of the MDCA-Network proposed in this article are as follows.

1) In this study, we introduce a cascade expansion attention architecture that employs depth-separable convolution to decouple modeling across various spectral bands. This is achieved by implementing a 7×7 large kernel convolution in conjunction with a 9×9 depth-convolution cascade structure, characterized by an expansion rate of 4, while also incorporating a channel grouping strategy. Furthermore, we propose a differentiable dynamic weight matrix that enables the automatic adjustment of the fusion ratio for multiscale features, tailored to the spectral response characteristics of the input data. This approach effectively addresses the issue of feature decoupling that arises from variations in spectral sensitivity within remote sensing images.
2) In this study, we present an algorithm for constructing virtual time series, aimed at generating a pseudo-time-series flow. This is accomplished by replicating the original features three times along the spectral dimension, utilizing bidirectional gated cyclic units to analyze the underlying patterns of feature evolution, and implementing a multiattention mechanism to create a feature correlation matrix across different years. To address the challenge of missing temporal features in monitoring interannual variations in static remote sensing images, we have developed a temporal consistency loss function. This function is specifically designed to guide the model in extracting light-invariant and seasonally robust representations of critical features.
3) In this study, the Indian Pines and Houston Hyperspectral datasets were initially selected to evaluate the generalizability of the classification model. Subsequently, a land use analysis was conducted for Rong'an County, located in Liuzhou City, Guangxi Zhuang Autonomous Region, which served as the focal area of the research. The analysis specifically examined changes in the distribution of forested areas and kumquat cultivation in Rong'an County from 2020 to 2024. The findings provide valuable data to support the ecological balance of the region, assess the economic significance of the kumquat industry, and inform planning for the sustainable utilization of land resources.

II. Research Methodology

This study employs pixel-level multispectral features as input, achieving land-cover classification through three stages:

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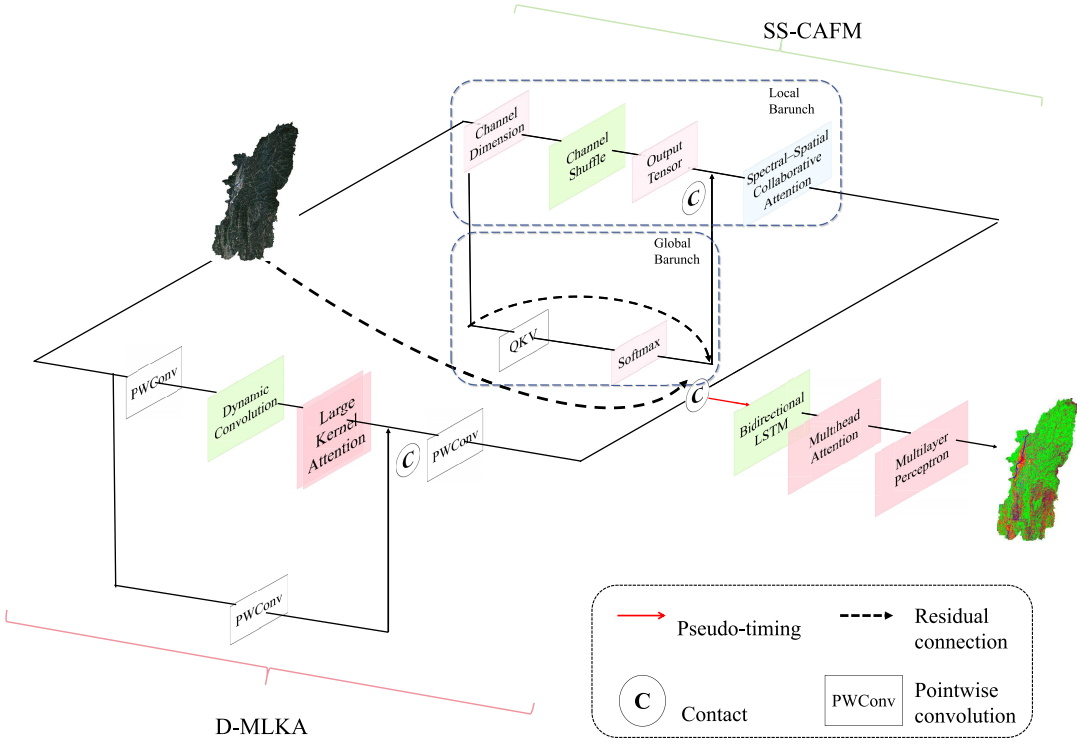


Fig. 1. Framework of the MDCA-Net model. The network receives multispectral pixel features as input. First, these features undergo parallel processing through two core modules: 1) the D-MLKA module described above; and 2) the SS-CAFM described below. The outputs from both modules are fused using a residual connection. The fused features are then reshaped and passed into a pseudo-temporal augmentation unit. Initially, a pseudo-temporal sequence is constructed through replication operations. Subsequently, a Bi-LSTM network extracts sequential contextual dependencies, and an MHA mechanism further refines the key information. The feature at the final time step is selected as the output. These high-level semantic features are ultimately classified into pixel-level object categories using an MLP classifier.

multiscale modeling, spectral–spatial dependency modeling, and pseudo-temporal enhancement. We independently model each pixel and design a progressive feature extraction pathway that progresses from spatial feature enhancement to spectral–spatial interaction, culminating in robust temporal enhancement. First, input features are simultaneously fed into two modules: D-MLKA and SS-CAFM. Their design objectives are complementary: D-MLKA focuses on an "adaptive multiscale receptive field," enhancing local structures and spectral variation patterns; while SS-CAFM emphasizes "spectral–spatial correlations and long-range dependencies," compensating for cross-channel and global connections that D-MLKA struggles to capture. Both modules process the same input in parallel, then fuse residual outputs to unify representations of local details, multiscale structures, and global spectral–spatial relationships. This architecture ensures functional synergy between modules rather than simple stacking. The fused features are reshaped and fed into a pseudo-temporal augmentation unit. By constructing fixed-length pseudo-temporal sequences, a bidirectional LSTM extracts sequential dependencies within the spectral dimension, while a multihead attention mechanism further highlights the most discriminative temporal steps. The final output is mapped via an MLP to classification probabilities, enabling pixel-level classification. The specific network architecture is illustrated in Fig. 1. Within the SS-CAFM framework, the spectral–spatial collaborative attention module is explicitly positioned as a module-level fusion unit, where globally attentive features and locally convolutional features are synergistically integrated. This design ensures that spectral dependencies and spatial structures are collaboratively modeled before temporal augmentation.

A. Dynamic Multiscale Large Kernel Attention

The proposed D-MLKA module extends the original MLKA [24] by introducing two key enhancements tailored for multispectral remote sensing imagery: 1) a learnable dynamic weighting mechanism for adaptive cross-scale feature fusion; and 2) a spectrum-aware channel grouping strategy that enables the network to automatically select appropriate receptive fields for different band subsets. Given an input feature map X, we first perform normalization and linear projection to obtain two branches, A and X'. The branch A is further split into two grouped subsets A_1 and A_2, each representing a distinct spectral band combination. Let A_1 and A_2 denote two channel-grouped subfeatures derived from the input feature A. Different convolution kernel sizes are applied to these subfeatures to capture multiscale contextual information. For each group, multiscale features are extracted using decomposed large-kernel depthwise convolutions

$$\mathrm{LKA}_k\left(A_j\right) = f_{\mathrm{PW}}\left(f_{\mathrm{DWD}}^{(k)}\left(f_{\mathrm{DW}}^{(k)}\left(A_j\right)\right)\right) \quad k \in \{5,7\}, \quad j \in \{1,2\}. \tag{1}$$

It should be noted that the subscripts 7 and 5 in $\mathrm{LKA}_7(A_i)$ and $\mathrm{LKA}_5(A_i)$ indicate the convolution kernel sizes, while A_1 and A_2 represent channel-grouped feature subsets. No additional feature tensors are introduced.

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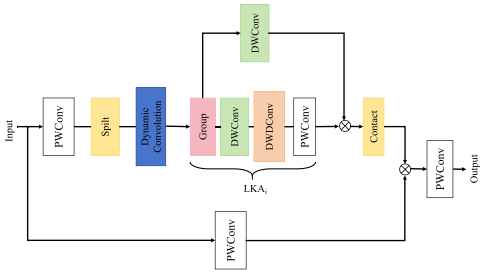


Fig. 2. Schematic of the D-MLKA architecture.

The 7×7 and 5×5 kernels provide distinct receptive fields for modeling objects of varying spatial scales. Unlike the original MLKA, where scale contributions are fixed, D-MLKA introduces a content-adaptive weighting scheme. To generate the dynamic weight parameters w_{dynamic}, we design a lightweight prediction network. Specifically, global average pooling is first applied to the input feature x to obtain a spectral response vector $g \in \mathbb{R}^C$. This vector is then fed into a two-layer MLP with GELU activation, compressing its dimension to 2 and producing the unnormalized weight vector $w_{\text{dynamic}} \in \mathbb{R}^2$. The final adaptive fusion weights are obtained via a Softmax function

$$\text{weights} = \text{Softmax}\left(w_{\text{dynamic}}\right), \quad w_{\text{dynamic}} \in \mathbb{R}^2 \tag{2}$$

which are generated from a lightweight network conditioned on the spectral response of the input. This allows the model to adaptively emphasize the most informative scale for each spectral group. The dynamically fused multiscale features are computed as

$$A_{\text{fused}} = \text{concat}\left(\text{weights}[0] \cdot \left(\text{LKA}_7\left(A_1\right) \odot f_{X7}\left(A_1\right)\right) \right. \\ \left. \text{weights}[1] \cdot \left(\text{LKA}_5\left(A_2\right) \odot f_{X5}\left(A_2\right)\right)\right). \tag{3}$$

The fusion feature A_{fused} is multiplied element-by-element with the projected X', projected by a 1×1 convolution, and output via explicit residual concatenation. Here, λ_{scale} is a learnable scaling factor that explicitly preserves the original input features

$$X_{\text{out}} = f_{\text{projiast}}\left(X' \otimes A_{\text{fused}}\right) \cdot \lambda_{\text{scale}} + X. \tag{4}$$

Through the combination of spectrum-aware channel grouping and dynamic multiscale weighting, D-MLKA achieves adaptive feature decoupling and selective receptive-field aggregation—two capabilities that the original MLKA lacks. This design effectively enhances the robustness of feature representation in the presence of heterogeneous band characteristics and spatial variability in remote sensing imagery. The specific structure of D-MLKA is shown in Fig. 2.

B. Spectral–Spatial Convolution and Attention Fusion Module

D-MLKA provides multiscale local enhancement but cannot model cross-channel spectral dependencies and long-range spatial relationships, thus requiring SS-CAFM to complement its shortcomings. To enhance the integration of global and local features across spectral and spatial dimensions in classification tasks, the SS-CAFM module is employed,

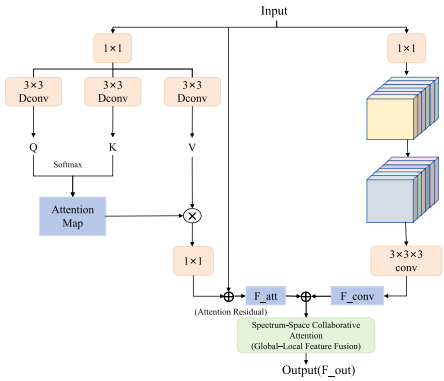


Fig. 3. SS-CAFM architecture schematic.

with its detailed architecture shown in Fig. 3. This module specifically addresses the challenge of spatial–spectral feature fusion in HSI classification. Its design is inspired by the effective use of the convolution and attention fusion module (CAFM) in denoising applications, which improves robustness against complex noise by jointly modeling spectral and spatial attention. The operational framework of the SS-CAFM module synergistically combines the local feature extraction capabilities of convolutional layers with the global dependency modeling provided by attention mechanisms [25]. Furthermore, it incorporates a lightweight spectral–spatial collaborative attention mechanism aimed at optimizing feature responses across both spectral channels and spatial dimensions.

First, the input features are processed in parallel by local branching and global branching. The local branching is done using Channel Shuffle with a 3-D convolution operation as shown in the following equation:

$$\mathbf{F}_{\text{conv}} = W_{3\times3\times3}\left(\text{CS}\left(W_{1\times1}\left(\mathbf{Y}\right)\right)\right). \tag{5}$$

where $W_{1\times1}$ and $W_{3\times3\times3}$ denote 1×1 convolution and $3\times3\times3$ convolution, respectively, and CS stands for channel shuffle operation for enhanced cross-channel interaction.

Global branching, on the other hand, captures long-range dependencies through an improved attention mechanism. In SS-CAFM, a new spectral–spatial co-attention module is added to compress the spectral channels by 1×1 convolution and generate spatial attention maps, $\mathbf{F}_{\text{att}}$ denotes global attention branching output

$$\mathbf{A}_{\text{ss}} = \text{Sigmoid}\left(W_{3\times3}\left(W_{1\times1}\left(\mathbf{F}_{\text{att}}\right)\right)\right). \tag{6}$$

$W_{1\times1}$ for channel dimension compression and $W_{3\times3}$ to generate spatial attention weights. The module fuses the original features with the attention-weighted features through residual linkage

$$\mathbf{F}_{\text{enhanced}} = \mathbf{F}_{\text{att}} \odot \mathbf{A}_{\text{ss}} + \mathbf{F}_{\text{att}}. \tag{7}$$

To explicitly model the collaborative interaction between spectral and spatial information, we introduce a spectral–spatial collaborative attention mechanism at the output stage of SS-CAFM. As illustrated in Fig. 3, the global attention-enhanced feature $\mathbf{F}_{\text{att}}$ encodes long-range spectral–spatial dependencies, while the local convolutional feature $\mathbf{F}_{\text{conv}}$ preserves fine-grained spatial structures

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such as edges and textures. The proposed collaborative attention mechanism integrates these two complementary representations through element-wise fusion, yielding the final output feature

$$\mathbf{F}_{\text{out}} = \mathbf{F}_{\text{att}} + \mathbf{F}_{\text{conv}}. \tag{8}$$

The process employed by SS-CAFM effectively characterizes the synergistic interplay between spectral and spatial dimensions while preserving the global–local feature fusion capabilities of CAFM. This approach not only enhances the attenuation of nonindependent and homogeneously distributed noise present in hyperspectral data but also reduces the potential decline in accuracy that may result from increased module complexity.

C. Bidirectional LSTM–MLP Joint Classification Architecture

After spatial feature extraction through the D-MLKA and SS-CAFM modules, the resulting feature representations are further enhanced via a pseudo-temporal modeling strategy based on Bi-LSTM networks. Although the input data correspond to single-temporal remote sensing imagery, the high-dimensional spectral features inherently exhibit ordered correlations along the spectral dimension. To exploit such intrinsic dependencies, the spectral features are reinterpreted as a fixed-length pseudo-temporal sequence.

Specifically, the fused feature vector is replicated along a newly introduced temporal axis to construct a stationary pseudo-temporal sequence with a length of three. This sequence length is empirically selected to balance contextual modeling capability and computational efficiency, as overly short sequences fail to provide sufficient dependency information, whereas longer sequences introduce redundancy and increase the risk of overfitting. The constructed pseudo-temporal sequence is then fed into a stacked bidirectional LSTM composed of two layers, enabling the model to capture bidirectional contextual dependencies within the spectral feature sequence.

At each pseudo-temporal step, the hidden states from the forward and backward LSTM branches are concatenated to form a comprehensive context-aware representation. To further emphasize the most informative temporal features for classification, a multihead self-attention (MHA) mechanism is applied to the Bi-LSTM outputs, which adaptively reweights the pseudo-temporal features according to their relevance. Only the attention-refined feature corresponding to the final pseudo-temporal step is retained as the high-level semantic representation.

Finally, the selected feature is passed through a lightweight MLP classifier consisting of linear layers, GELU activation, Layer Normalization, and Dropout regularization. The MLP performs nonlinear feature transformation and regularization, and the output is projected into the category space through a linear classifier, completing the pixel-level land-cover classification. The overall Bi-LSTM–MLP joint classification framework is illustrated in Fig. 4.

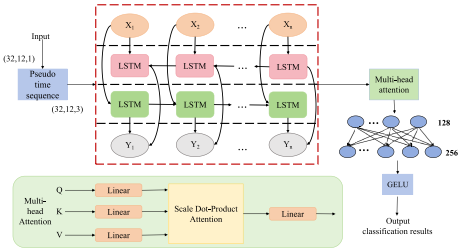


Fig. 4. Bidirectional LSTM–MLP joint classification architecture diagram.

In conclusion, the MDCA model focuses on pixel-level classification of remote sensing images as its primary objective. Initially, multispectral pixel features are integrated at multiple scales through the spatial feature extraction modules, D-MLKA and SS-CAFM. The D-MLKA module employs dynamic weights to adjust the contributions of convolutional kernels across various scales while preserving the original features through residual concatenation. Simultaneously, the SS-CAFM module enhances cross-channel feature interactions by combining global attention mechanisms with local convolutional joint modeling. Subsequently, the features are transformed into a pseudo-temporal sequence, which captures temporal context dependencies using bidirectional long short-term memory networks. Key time-step information is filtered through a multihead attention mechanism, ultimately leading to the extraction of high-level semantic features from the final frame. These features undergo nonlinear transformation and regularization via an MLP, culminating in the output of the precise feature class through a linear classifier.

III. Experimental Results

A. Description of the Dataset

To evaluate the accuracy of the model presented in this article, as well as the reliability of the qualitative findings and the general applicability of the algorithm for land-cover classification using multisource satellite remote sensing imagery, we selected two authoritative public hyperspectral datasets: the Houston dataset and the Indian Pines dataset. These datasets were utilized for systematic evaluation in our experiments. Additionally, this study incorporated the 2024 Rong'an County dataset, derived from Sentinel-2 satellite imagery obtained from the European Space Agency (ESA) data repository. After acquiring the data, we resampled the bands to a resolution of 10 m using Snap 8.0 software. Subsequently, we synthesized the 12 bands of the Sentinel-2 imagery sequentially using ENVI 5.6 software, employed the seamless mosaic tool to facilitate image stitching, performed image cropping, and ultimately exported the processed image data of Guangxi Rong'an County in TIFF format.

1) Houston University Dataset: The dataset, obtained from aerial remote sensing observations conducted by the University of Houston, Texas, using the CASI-1500 hyperspectral sensor developed by ITRES, has been made publicly available by the National Center for Airborne Laser Mapping (NCALM), which is supported by funding from the U.S. National Science Foundation (NSF). This imagery covers a spatial extent of 349 × 1905 pixels and consists of 144 contiguous spectral

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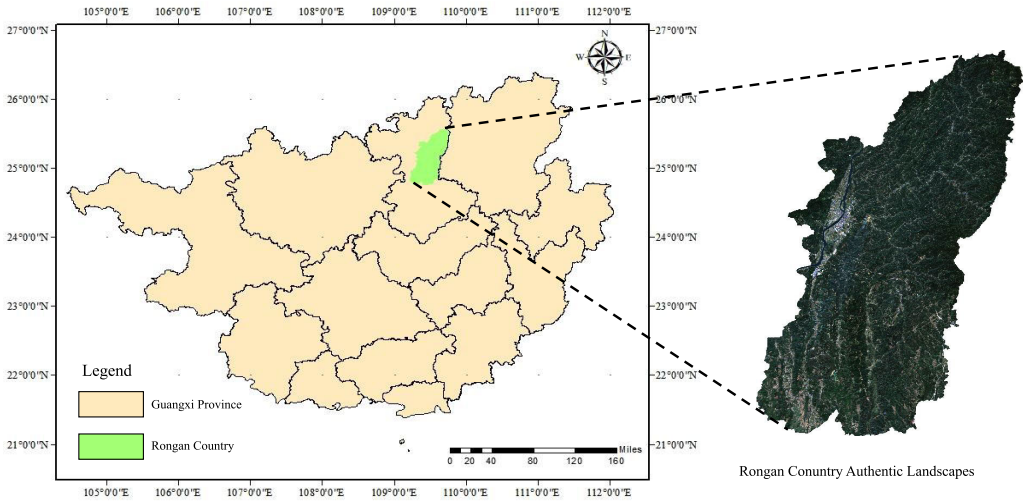


Fig. 5. Geographic location of Rong'an and corresponding remote sensing images.

bands within the range of 364–1046 nm, achieving a spatial resolution of 2.5 m.

2) Indian Pines Dataset: The Indian Pines dataset was acquired in 1992 using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), a technology developed by NASA's Jet Propulsion Laboratory (JPL). This dataset focuses on the Indian Pines Experimental Area (IPEA), located in northwestern Indiana. The original image consists of 145 × 145 pixels and has a spatial resolution of 20 m, covering a broad spectral range from 400 to 2500 nm, with a total of 220 spectral bands. To improve data quality, bands 104–108, 150–163, and 220, which are affected by water vapor absorption and sensor noise, have been systematically excluded from this analysis, resulting in the retention of 200 effective bands for further examination.

3) Rong'an Dataset: Rong'an County, administered by Liuzhou City in the Guangxi Zhuang Autonomous Region, is located in the northern part of this autonomous area, situated between latitudes 24°46' and 25°34' North, and longitudes 109°13' and 109°47' East. This country lies North of the Tropic of Cancer and South of the North Temperate Zone. It extends 89 km in length from North to South and 44.5 km in width from East to West, covering an administrative area of approximately 2901.28 km^2. Rong'an County is characterized by a central subtropical monsoon climate, which features intense solar radiation, a warm temperature profile, abundant rainfall, a brief winter season, and an extended summer season, with precipitation and heat occurring simultaneously. This climatic condition contributes to the region's rich agricultural resources. Notably, Rong'an County is recognized as a significant kumquat production area in China, with the kumquat industry playing a crucial role in the local economy. Accurately assessing the kumquat cultivation area is essential for promoting the growth of this industry and achieving precise yield estimations, which are vital for enhancing management practices within the sector and ensuring the agricultural benefits of the region. The specific geographic information pertaining to the study area is illustrated in Fig. 5.

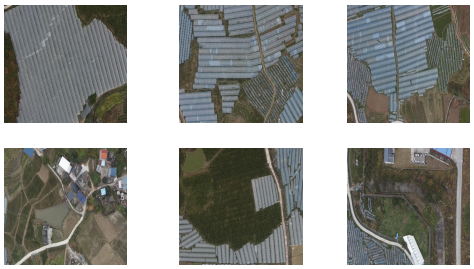


Fig. 6. Drones collecting images on the ground in Rong'an County.

In this research, seven distinct feature types—namely buildings, water bodies, kumquats, trees, ponds, bare ground, and other land—were examined using DJI AAVs equipped with multispectral sensors to conduct low-altitude aerial photography in designated key areas. The study involved the preestablishment of GPS coordinate points within pure sample areas, facilitating a data collection methodology that integrated fixed-point hovering photography with gridded aerial strip scanning to yield high-precision surface spectral feature data. Subsequently, the AAV imagery was aligned with the Sentinel-2 satellite remote sensing base map at the pixel level, utilizing the region of interest (ROI) tool in ENVI 5.6 software for visual annotation, which ultimately led to the creation of a sample library for Rong'an County. A total of 12 867 sample points representing various feature types were collected. During the field collection of kumquat and other samples, emphasis was placed on selecting contiguous planting areas exceeding 100 m^2 to enhance the accumulation of prior knowledge and facilitate subsequent accuracy verification. To ensure the model has sufficient data for effective learning and prevent overfitting caused by insufficient training samples, thereby obtaining more stable and reliable model parameters, 80% of the samples from each category are randomly selected as the training set, with the remaining 20% serving as the test set. The specific AAV-acquired images are illustrated in Fig. 6. The distribution of specific category labels and sample sizes in the study area is detailed in the statistics in Table I.

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TABLE I
CATEGORY INFORMATION FOR THE DATASET USED

Dataset Name	ID	Category	Train	Test	ID	Category	Train	Test
Houston	1	Healthy Grass	198	1053	9	Road	193	1059
	2	Stressed Grass	190	1064	10	Highway	191	1036
	3	Synthetic Grass	192	505	11	Railway	181	1054
	4	Tree	188	1056	12	Parking Lot1	192	1041
	5	Soil	186	1056	13	Parking Lot2	184	285
	6	Water	182	143	14	Tennis Court	181	247
	7	Residential	196	1072	15	Running Track	187	473
	8	Commercial	191	1053				
Indian Pines	1	Corn Notill	50	1384	9	Soybean Clean	50	564
	2	Corn Mintill	50	784	10	Wheat	50	162
	3	Corn	50	184	11	Woods	50	1244
	4	Grass Pasture	50	447	12	Buildings Grass Trees Drives	50	330
	5	Grass Trees	50	697	13	Stones Steel Towers	50	45
	6	Hay Windrowed	50	439	14	Alfalfa	15	39
	7	Soybean Notill	50	918	15	Grass Pasture Mowed	15	11
	8	Soybean Mintill	50	2418	16	Oats	15	5
Rongan	1	Buildup	2488	623				
	2	River	1900	475				
	3	Kumquat	1340	336				
	4	Tree	1196	299				
	5	Pond	1327	332				
	6	Barren	1065	353				
	7	Other	906	227				

B. Experimental Setup

In this research, the coordinate data related to the area of interest are extracted using the ENVI 5.6 platform. The resulting spatial coordinate dataset is then formatted as a text file to serve as the training sample set for DL applications. The model's parameter configurations include a batch size of 32 and a learning rate decay factor of 0.9, along with a dynamic adjustment strategy designed to enhance the training process. The algorithm is implemented in a Python 3.7 programming environment, utilizing the PyTorch 1.12.0 DL framework. The hardware setup consists of a Windows 10 operating system, a 12th-generation Intel Core i5-8500 processor, and an NVIDIA GeForce RTX 3060 graphics card, with optimizations for parallel computation achieved through CUDA acceleration.

In this study, three evaluation metrics derived from confusion matrices—overall accuracy (OA), average accuracy (AA), and the Kappa coefficient—are utilized to assess the classification performance of various models in the context of multispectral pixel classification. The cross-entropy loss function is chosen for this task, as it enhances the model's ability to distinguish between different categories in remote sensing image classification. Additionally, this loss function can be minimized using optimization techniques, such as gradient descent, thereby improving the model's accuracy.

C. Ablation Experiments

To evaluate the performance of the network structure proposed in this article, ablation experiments are first conducted on the dataset of the Indian Pines study area. The results of the specific experimental experiments are shown in Table II.

The OA of the joint architecture network, which combines BiLSTM and MLP, is recorded at 67.16%. The independent integration of the MLKA and CAFM leads to a general enhancement in image classification accuracy, with MLKA exhibiting a more pronounced effect, achieving an OA of 71.90%. This improvement can be attributed to MLKA's ability to directly address the imbalance between the multiscale sensitivity of spectral features and the local-global context dependency inherent in pixel-level classification of remote sensing images through its multiscale mechanism. In contrast, while CAFM enhances feature interaction by integrating attention mechanisms with local convolution, its global attention struggles to effectively model long-range dependencies in single-pixel inputs. Furthermore, the local convolution is reduced to a linear transformation of the channel dimension due to insufficient spatial neighborhood information, which diminishes its capacity to extract multiscale characteristics of spectral features compared to MLKA, thereby hindering significant accuracy improvements. The testing time gradually increases with the addition and improvement of MLKA and CAFM modules.

Upon the individual implementation of the D-MLKA and the SS-CAFM, the OA is elevated to 75.38% and 71.37%, respectively, reflecting increases of 3.48% and 3.96% over the previous metrics. The enhanced performance of D-MLKA is attributed to its dynamic adjustment of the fusion ratio in multiscale large kernel convolution, which mitigates feature redundancy and information loss associated with traditional

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TABLE II
RESULTS OF ABLATION EXPERIMENTS OF MDCA-NET WITH DIFFERENT MODULE COMBINATIONS ON INDIAN PINES STUDY AREA

No.	Model Variant	Different Module				Metric			Test Time(s)
		MLKA	D-MLKA	CAFM	SS-CAFM	OA(%)	AA(%)	Kappa	
1	Baseline (BiLSTM + MLP)	×	×	×	×	67.16	68.66	0.6311	94.11
2	Baseline + MLKA	✓	×	×	×	71.90	80.63	0.6820	164.71
3	Baseline + D-MLKA	×	✓	×	×	75.38	84.36	0.7212	167.55
4	Baseline + CAFM	×	×	✓	×	67.41	74.12	0.6317	163.98
5	Baseline + SS-CAFM	×	×	×	✓	71.37	78.56	0.6757	185.16
6	Baseline + MLKA + CAFM	✓	×	✓	×	75.63	81.46	0.7223	232.82
7	Baseline + MLKA + SS-CAFM	✓	×	×	✓	76.06	84.26	0.7285	251.16
8	Baseline + D-MLKA + CAFM	×	✓	✓	×	75.48	83.70	0.7213	245.55
9	MDCA-NET (NO Grouping)	×	✓	×	✓	73.99	83.26	0.7054	209.97
10	MDCA-NET(NO Adaptive Fusion)	×	✓	×	✓	70.89	81.03	0.7391	232.61
11	MDCA-NET	×	✓	×	✓	77.38↑	84.69↑	0.7428↑	261.46

Note: Each row represents a distinct ablation configuration constructed by selectively enabling different modules within the proposed framework.

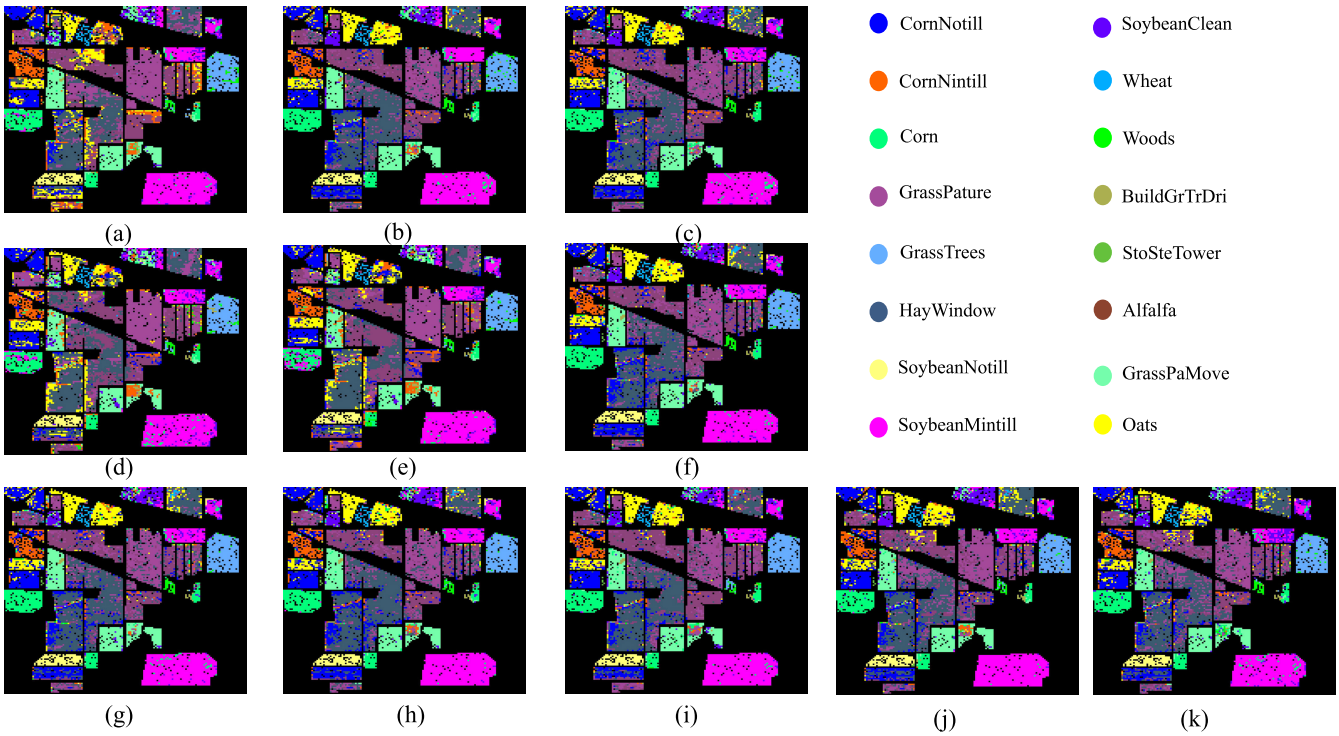


Fig. 7. Classification results of MDCA-NET using different module combinations on the dataset of the Indian Pines study area. (a) BiLSTM+MLP. (b) MDCA-NET (only MLKA). (c) MDCA-NET (only D-MLKA). (d) MDCA-NET (only CAFM). (e) MDCA-NET (only SS-CAFM). (f) MDCA-NET (MLKA+CAFM). (g) MDCA-NET (MLKA+SS-CAFM). (h) MDCA-NET (D-MLKA+CAFM). (i) MDCA-NET (D-MLKA+SS-CAFM). (j) MDCA-NET (no grouping). (k) MDCA-NET (no adaptive fusion).

fixed multiscale designs. This adjustment directly improves the discriminative capacity of the classification boundary region. In contrast, SS-CAFM improves classification performance by jointly modeling spatial and spectral dependencies through the integration of static local convolution and attention-based feature reweighting. The channel-shuffled convolution branch preserves fine-grained local spatial structures, while the spectral–spatial attention branch emphasizes informative spectral responses and suppresses redundant or noisy features. By compensating for the loss of spatial context caused by pixel-wise input through localized convolution and attention-guided feature enhancement, SS-CAFM effectively improves robustness against spectral overlap in complex land-cover scenarios. Subsequently, the two sets of modules are integrated, leading to further experimentation. The finalized MDCA network achieved an OA of 77.38% and an AA of 84.69%, representing improvements of 1.75% and 3.23%, respectively, compared to the integration of unmodified MLKA and CAFM modules. The kappa coefficient reached 0.7428, demonstrating the robust effectiveness of each component in pixel-level classification of remote sensing imagery. To further validate the proposed feature decoupling mechanism, ablation studies were conducted by systematically removing the channel grouping strategy and the dynamic allocation mechanism. The overall performance decreased by 3.39% and 6.49%, respectively, highlighting the importance of these components. The specific categorization is shown in Fig. 7.

The length T of the pseudo-time sequence is a critical design choice for the model. To validate its impact and determine the optimal value, we conducted comparative experiments on

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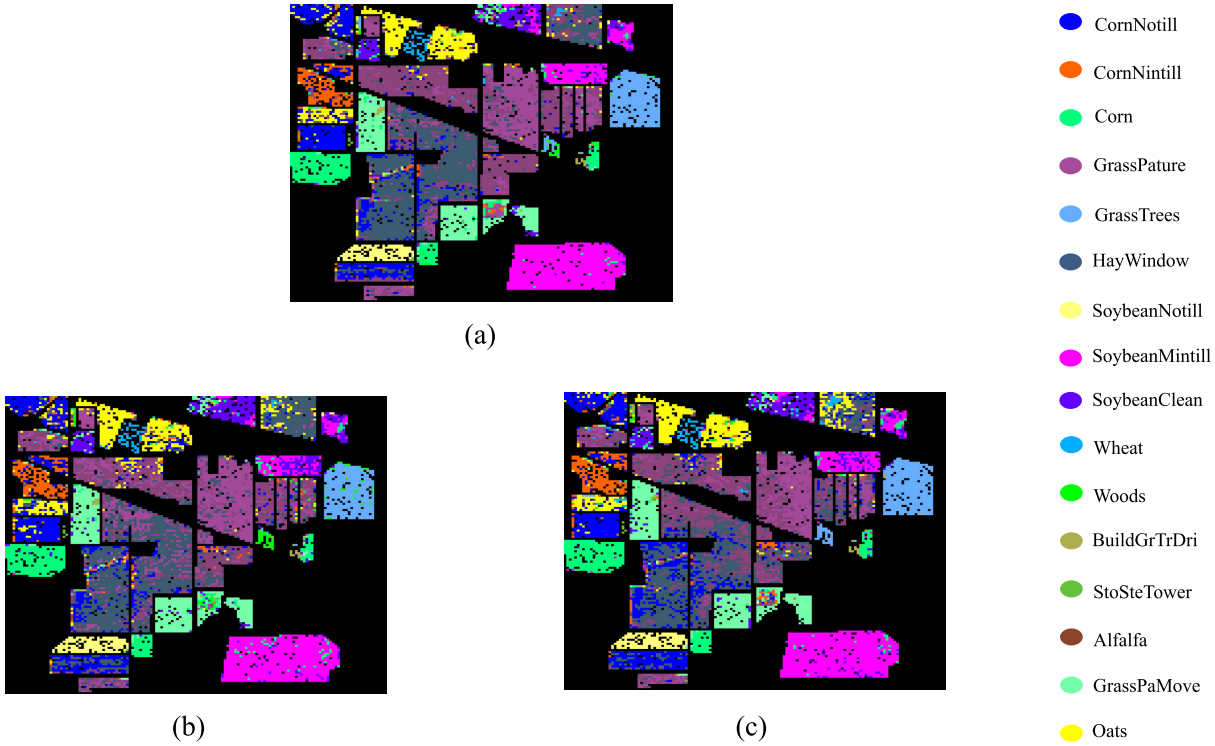


Fig. 8. Classification results of the Indian Pines study area dataset using MDCA with different pseudo-time sequence lengths. (a) T = 3. (b) T = 1. (c) T = 5.

TABLE III

ABLATION TEST RESULTS OF MDCA-NET WITH DIFFERENT PSEUDO-TIME-SERIES LENGTHS IN THE INDIAN PINES STUDY AREA

No.	Different Methods	Different Module				Metric			Test Time(s)
		MLKA	D-MLKA	CAFM	SS-CAFM	OA(%)	AA(%)	Kappa	
1	MDCA-NET(T=3)	×	✓	×	✓	77.38	84.69	0.7428	261.46
2	MDCA-NET(T=1)	×	✓	×	✓	73.14	82.82	0.6950	235.66
3	MDCA-NET(T=5)	×	✓	×	✓	73.68	83.26	0.7008	235.83

the Indian Pines dataset by controlling all other variables and varying only the value of T. The experimental results are shown in Table III.

At T = 1, the model exhibits the lowest performance because a single static feature point does not provide sufficient sequential context for effective Bi-LSTM modeling. When T is set to 3, all key performance metrics reach their maximum values. However, increasing T to 5 leads to a decline in performance, suggesting that excessively long sequences may introduce noise or redundancy, thereby hindering model optimization. In conclusion, T = 3 achieves an optimal balance between model performance and computational efficiency. This experiment strongly supports our design choice to construct pseudo-time sequences of length three. Detailed classification results are presented in Fig. 8.

D. Comparative Experimental Results

This experiment employs traditional algorithms SVM [26] and KNN [27], alongside DL algorithms VIT [15], CAF [16], Morphformer [17], SSFTT [19], and CACFT [20] as comparison methods. The SVM algorithm was configured with a radial basis function (RBF) kernel and a penalty factor of 10, establishing classification boundaries in "one-versus-all" (ovr) mode. The KNN algorithm employed a voting decision mechanism based on the three nearest neighbor samples.

To evaluate the universality and generalization capabilities of the classification model presented in this study for land-cover classification utilizing multisource satellite remote sensing imagery, we selected representative hyperspectral datasets, specifically the Houston and Indian Pines datasets, for comparative analysis. As demonstrated in Tables IV and V, a comparison of the quantitative metrics of the MDCA-NET model against other leading models indicates that MDCA-NET exhibits significant advantages on the Indian Pines dataset, achieving an OA of 77.03%, an AA of 85.31%, and a Kappa coefficient of 0.7391, outperforming alternative methods, particularly in the classification of crop categories such as CornNotill and HayWindow, where notable improvements in accuracy are observed. Nevertheless, MDCA-NET excelled in classifying urban features, such as Road and parkinglot1, within the Houston dataset, attaining an AA of 85.32%. This finding underscores the model's adaptability to varying feature characteristics. Qualitative visualizations presented in Figs. 9 and 10 further illustrate that MDCA-NET maintains superior continuity of classified patches across both datasets while significantly reducing misclassification noise, thereby validating

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TABLE IV
COMPARISON OF CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON INDIAN PINES STUDY AREA

ClassNo.	Different Methods							
	SVM	KNN	VIT	CAF	Morphformer	SSFTT	CACFT	MDCA-NET
1	65.82	56.14	38.65	54.69	75.00	55.99	62.78	63.72
2	54.84	53.69	39.28	76.65	72.95	86.22	69.13	78.69
3	92.93	77.17	93.47	85.32	83.69	84.23	83.15	85.86
4	92.39	89.71	78.29	91.27	93.95	93.06	91.05	89.93
5	85.50	77.33	71.87	84.50	93.68	87.37	87.66	87.66
6	98.40	94.98	92.71	93.84	89.74	92.93	92.71	97.03
7	80.50	62.96	62.20	83.98	81.91	71.45	83.44	72.33
8	54.75	43.67	68.03	67.49	53.26	62.94	67.41	67.82
9	73.04	45.04	26.59	67.02	70.56	70.39	73.22	76.77
10	98.14	94.44	96.29	98.76	99.38	99.38	93.20	98.14
11	93.56	73.55	66.47	88.26	90.43	87.54	90.99	94.05
12	52.42	35.15	24.54	72.42	67.27	84.84	60.00	69.39
13	97.77	97.77	97.77	100.0	97.77	88.88	97.77	95.55
14	33.33	79.49	25.64	84.61	87.17	87.17	84.61	87.17
15	90.90	81.81	81.81	90.90	100.00	100.00	90.90	90.90
16	100.00	80.00	40.00	80.00	100.00	60.00	100.00	100.00
OA(%)	72.34	60.56	59.64	75.42	75.18	74.78	76.22	77.38↑
AA(%)	79.02	71.43	62.73	82.49	84.80	82.03	83.01	84.69↑
Kappa	0.6886	0.5564	0.5371	0.7221	0.7197	0.7145	0.7301	0.7428↑
Test Time(s)	3.43	4.71	252.44	328.13	357.93	122.43	334.54	261.46

TABLE V
COMPARISON OF CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON HOUSTON STUDY AREA

ClassNo.	Different Methods							
	SVM	KNN	VIT	CAF	Morphformer	SSFTT	CACFT	MDCA-NET
1	83.00	83.19	83.47	83.00	85.28	88.31	83.47	84.42
2	97.93	96.14	96.80	94.73	95.86	96.61	93.98	95.77
3	99.60	99.60	99.40	100.00	99.80	99.80	100.00	99.80
4	98.39	98.29	97.53	99.33	92.80	96.49	99.14	96.11
5	96.30	96.68	97.44	98.10	95.26	96.21	95.64	98.20
6	95.10	94.40	95.10	99.30	95.10	95.10	95.10	98.60
7	87.40	83.58	76.30	79.85	60.44	89.27	79.94	81.71
8	42.07	48.90	53.75	53.37	69.61	61.63	59.44	62.01
9	80.16	69.68	64.11	71.01	67.98	74.59	58.54	80.54
10	48.84	70.46	41.02	65.44	66.69	55.98	67.56	55.21
11	87.66	81.50	79.79	69.82	96.86	77.89	80.26	85.00
12	58.30	50.62	52.92	60.32	72.81	77.13	54.75	77.61
13	42.45	41.75	48.07	62.45	64.91	62.10	58.24	69.12
14	98.38	98.38	99.19	98.78	99.19	99.59	99.59	97.57
15	98.09	98.09	98.09	97.88	98.09	98.30	98.73	98.09
OA(%)	79.57	79.42	76.51	79.62	82.06	82.98	79.32	83.34↑
AA(%)	80.92	80.76	78.87	82.23	84.05	84.61	81.63	85.32↑
Kappa	0.7784	0.7769	0.7454	0.7791	0.8053	0.8153	0.7759	0.8194↑
Test Time(s)	8.18	9.19	702.94	922.83	971.77	324.58	896.71	664.92

the effectiveness of its integrated spatial–spectral modeling approach. The experimental results suggest that the proposed method effectively balances classification accuracy with scene generalization capabilities in hyperspectral contexts.

The quantitative categorization outcomes for the three composite metrics—OA, AA, and Kappa coefficient—related to the dataset from the Rong'an County study area are presented in Table VI. The results demonstrate that the MDCA model

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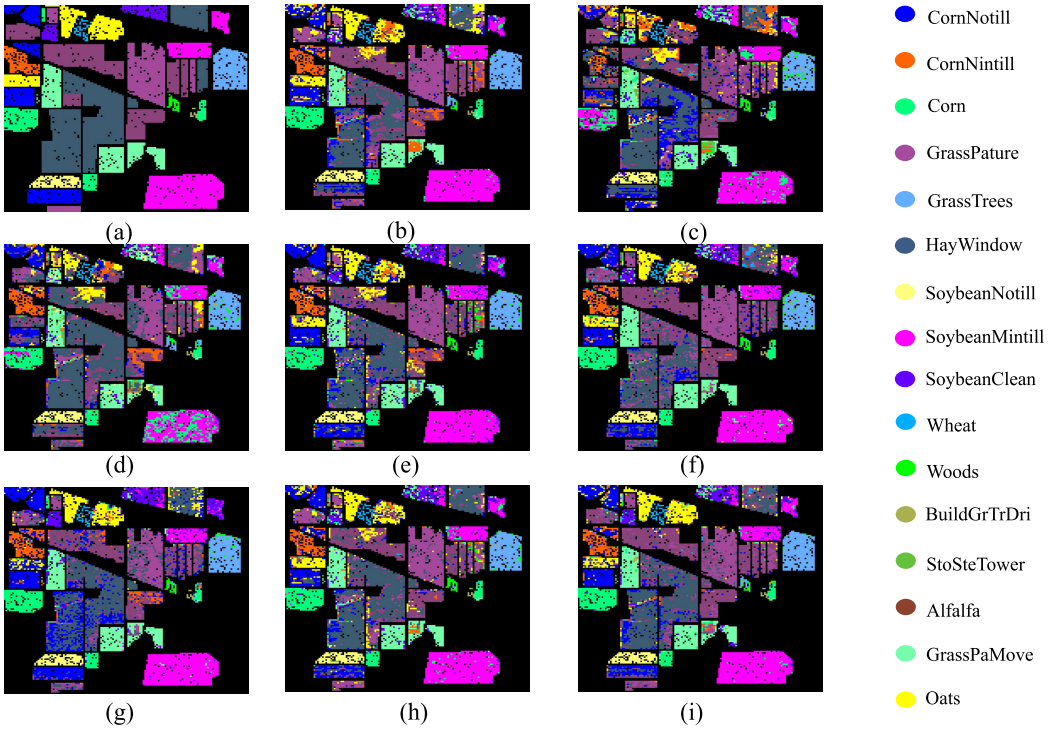


Fig. 9. Classification results obtained by different models on Indian pines data. (a) Ground truth. (b) SVM. (c) KNN. (d) VIT. (e) CAF. (f) Morphformer. (g) SSFTT. (h) CACFT. (i) MDCA-NET.

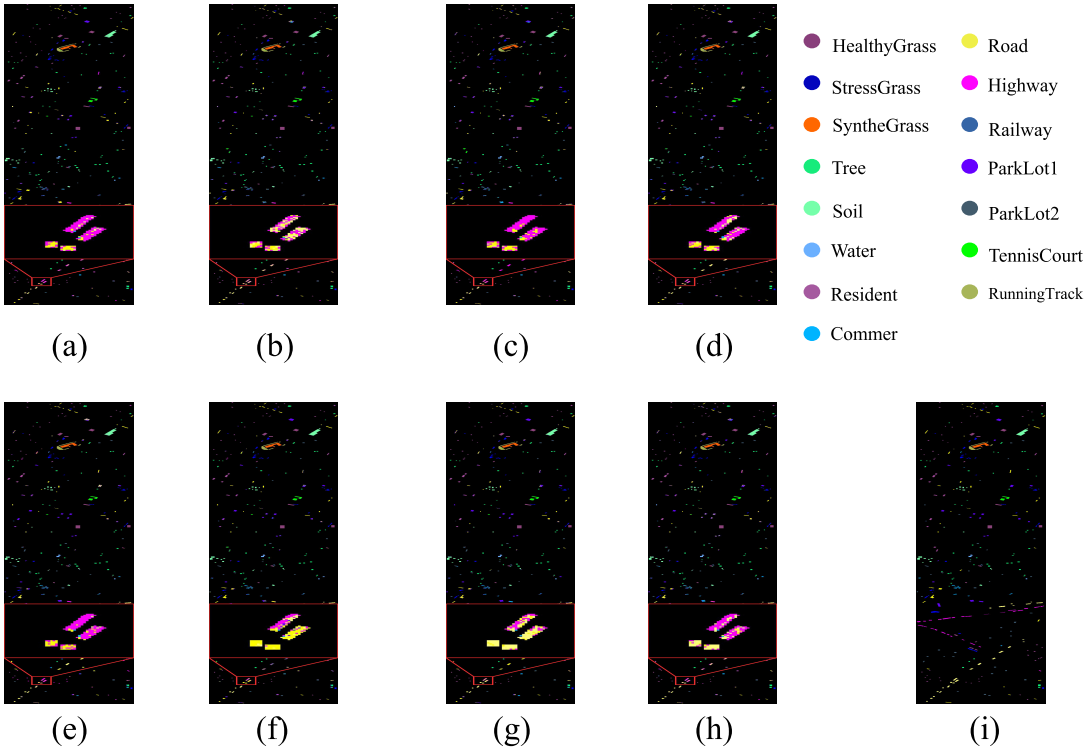


Fig. 10. Classification results obtained by different models on Houston data. (a) SVM. (b) KNN. (c) VIT. (d) CAF. (e) Morphformer. (f) SSFTT. (g) CACFT. (h) MDCA-NET. (i) Ground truth.

proposed in this study surpasses other models, achieving OA, AA, and Kappa values of 98.66%, 98.43%, and 0.9845, respectively. In contrast, traditional methods such as SVM and KNN yield OA values of 91.18% and 94.40%, respectively. This discrepancy can be attributed to their reliance on manual feature extraction, which leads to suboptimal performance in the "Other" category. These findings highlight the advantages of DL models in facilitating automatic feature learning and modeling nonlinear relationships.

The second-tier models, CAF and VIT, improved classification performance by employing attention mechanisms and multiscale feature fusion, achieving OA of 97.68% and

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TABLE VI
COMPARISON OF CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE RONG'AN COUNTY 2024 DATASET

Class	Different Methods							
	SVM	KNN	VIT	CAF	Morphformer	SSFTT	CACFT	MDCA-NET
Buildup	85.07	88.12	98.00	97.14	98.60	94.89	96.66	99.08
River	93.89	97.05	98.73	99.57	97.32	98.65	97.18	99.71
Kumquat	85.11	93.15	93.60	95.70	91.92	95.91	94.51	96.21
Tree	100.00	100.00	100.00	100.00	100.00	99.88	100.00	100.00
Pond	98.19	99.69	97.17	96.96	99.49	99.79	98.79	100.00
Barren	87.53	91.78	97.43	97.62	95.62	96.58	97.81	98.19
Other	84.14	89.86	92.03	93.21	83.48	86.09	84.74	94.23
OA(%)	91.18	94.40	97.39	97.68	96.47	96.70	96.62	98.66↑
AA(%)	91.74	94.46	97.21	97.53	95.81	96.48	96.21	98.43↑
Kappa	0.8981	0.9352	0.9697	0.9730	0.9590	0.9618	0.9608	0.9845↑

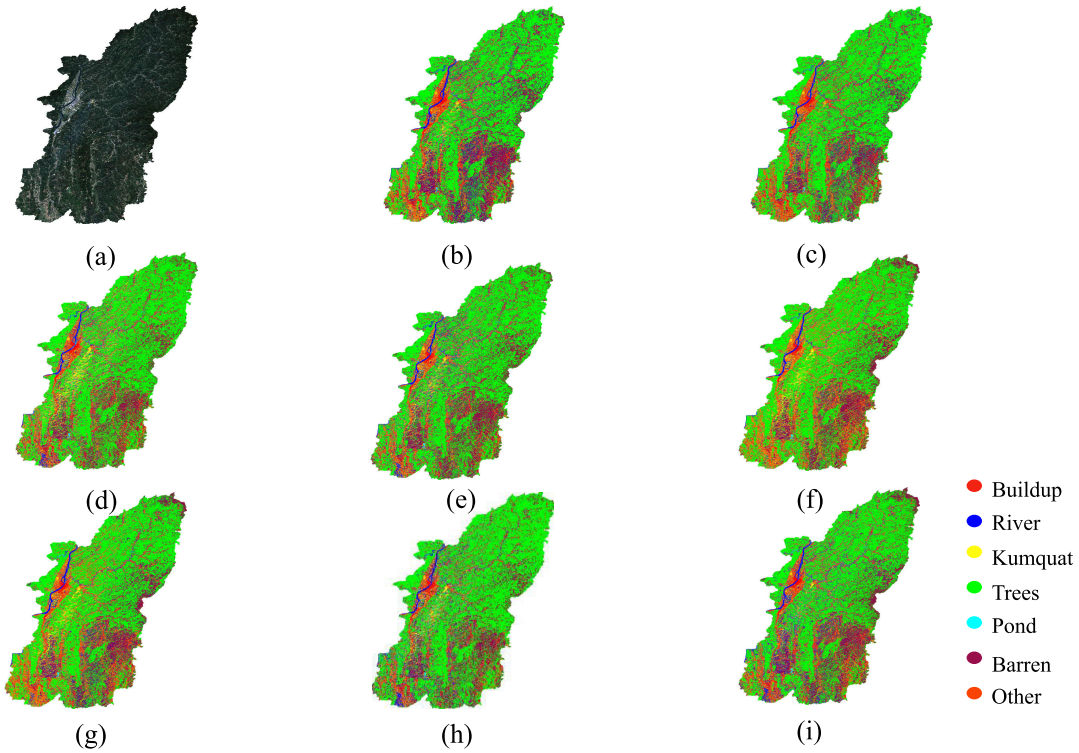


Fig. 11. Classification results obtained by different models on the Rong'an County 2024 dataset. (a) Image. (b) SVM. (c) KNN. (d) VIT. (e) CAF. (f) Morphformer. (g) SSFTT. (h) CACFT. (i) MDCA-NET.

97.39%, respectively. However, their performance remained significantly inferior to that of MDCA-NET. Within the Other category, characterized by complex features, SSFTT and CACFT attained accuracies of only 86.09% and 84.74%, respectively. In contrast, MDCA-NET substantially outperformed these models, achieving 94.23% accuracy and surpassing the second-best model, CAF (93.21%), by 1.02%. This result confirms the adaptability of MDCA-NET's dynamic attention mechanism to intricate scenarios. Furthermore, in the Barren and Buildup categories, MDCA-NET demonstrated superior performance, with overall accuracies (OAs) of 98.19% and 99.08%, respectively, compared to CACFT's 97.62% and 97.14%. These findings highlight the effectiveness of multiscale feature fusion in handling texturally complex objects. Taking into account the two datasets, KNN and SVM have the shortest testing time, while SSFTT has the highest efficiency in the DL network architecture. This is due to the highly optimized Transformer structure, which greatly reduces the computational complexity of the self-attention mechanism. Next is MDCA, which introduces multiscale convolution and an attention mechanism, which brings certain computational overhead. However, through lightweight design such as depthwise separable convolution and dynamic weights, it effectively controls the computational burden while ensuring performance. A detailed comparative classification diagram is presented in Fig. 11.

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TABLE VII
COMPARISON OF CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE RONG'AN COUNTY 2022 DATASET

Class	Different Methods							
	SVM	KNN	VIT	CAF	Morphformer	SSFTT	CACFT	MDCA-NET
Buildup	85.39	89.08	98.00	96.44	98.11	96.61	94.08	97.63
River	96.21	97.68	98.73	96.97	98.38	96.96	96.55	99.57
Kumquat	91.66	94.04	93.60	92.40	98.00	94.11	96.31	96.51
Tree	96.65	97.99	100.00	93.41	98.66	96.98	94.84	99.88
Pond	96.68	97.59	97.17	97.37	99.39	98.08	98.58	98.98
Barren	87.53	90.65	97.73	90.86	92.57	92.59	97.71	97.81
Other	74.88	80.61	92.03	83.48	91.15	86.68	85.92	94.52
OA(%)	91.15	93.51	97.39	94.53	97.32	95.66	95.71	98.21↑
AA(%)	91.13	93.46	97.12	93.87	97.04	95.26	95.50	98.12↑
Kappa	0.8975	0.9247	0.9697	0.9365	0.9689	0.9497	0.9502	0.9793↑

TABLE VIII
COMPARISON OF CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE RONG'AN COUNTY 2020 DATASET

Class	Different Methods							
	SVM	KNN	VIT	CAF	Morphformer	SSFTT	CACFT	MDCA-NET
Buildup	89.88	91.33	97.25	96.66	97.74	95.96	97.20	98.97
River	96.90	98.83	97.27	98.83	99.87	99.74	99.02	100.00
Kumquat	87.20	88.09	95.01	94.72	98.20	95.91	89.91	95.30
Tree	98.48	97.97	98.81	98.30	97.63	95.43	98.22	99.40
Pond	62.95	96.08	95.57	96.68	97.98	98.99	96.27	99.59
Barren	60.62	80.45	88.89	91.73	94.67	93.43	88.11	96.57
Other	66.51	91.49	80.97	80.82	85.18	76.69	86.26	92.47
OA(%)	84.79	92.43	95.12	95.60	97.09	95.51	95.15	98.21↑
AA(%)	82.82	91.78	94.23	94.72	96.41	94.52	94.38	97.79↑
Kappa	0.8233	0.9122	0.9434	0.9489	0.9663	0.9479	0.9438	0.9792↑

E. Vegetation Dynamics

This study employs the Sentinel-2 image series acquired from the ESA website, specifically focusing on the dates of December 21, 2020, and December 12, 2022, to examine the dynamics of feature classification in Rong'an County. Following the preprocessing phase, labels were generated using 2024 labeled ROIs based on prior knowledge, and a text file containing the coordinates of these ROIs was created. The sample set was subsequently divided randomly into training and test sets in a 60:40 ratio, as specified by the coordinates in the text file. The proposed MDCA method was then utilized to assess changes in vegetation utilization within the study area, with a particular focus on the variations in the areas of Kumquat and Trees.

The classification outcomes of various models applied to the datasets from 2020 and 2022 are presented in Tables VII and VIII, which encompass data spanning three years. The summarized results of the MDCA model indicate superior performance in classifying data from both years, achieving the highest accuracy among all models evaluated. This finding confirms the robustness of its architecture in cross-year and cross-scene applications. Traditional methods, such as the SVM and KNN, consistently demonstrate inferior performance. Specifically, the SVM recorded an accuracy of only 66.51% in the Other category in 2020, which improved to 74.88% in 2022; however, this remained significantly lower than the accuracy of MDCA-NET, which ranged from 92.47% to 94.52% during the same period. Models utilizing attention mechanisms, including the ViT, Morphformer, and CAF, exhibit notable performance enhancements through global feature fusion; nevertheless, they still do not match the performance of MDCA-NET, with CAF achieving an overall accuracy of 95.60% in 2020 but declining to 94.53% in 2022. In contrast, MDCA-NET maintains a stable overall accuracy of 98% or higher across the three years, thereby underscoring its generalization capabilities.

The classification of trees and rivers demonstrates a consistent stability across most years; however, certain categories exhibit sensitivity to temporal variations. Notably, the Kumquat category achieves a three-year accuracy ranging from 95.30% to 96.51% when utilizing the MDCA-NET model. The Other category shows a year-on-year improvement in accuracy with MDCA-NET, significantly outperforming other models during the same period, thereby substantiating the

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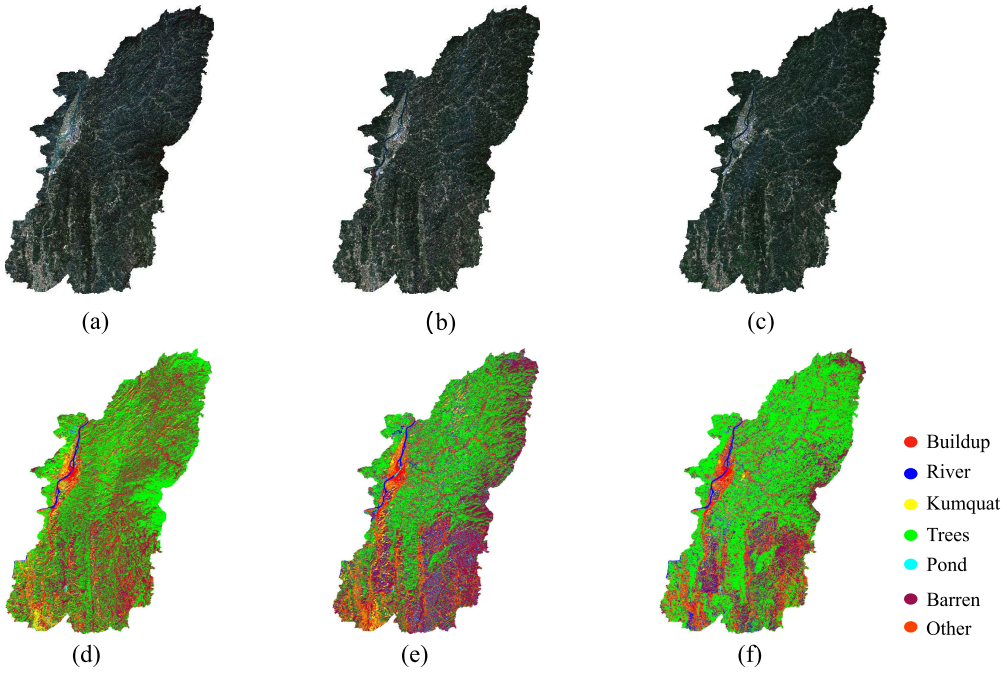


Fig. 12. True-color maps of three-phase imagery. (a) 2020, (b) 2022, and (c) 2024. Spatial distribution of features in three-phase imagery: (d) 2020, (e) 2022, and (f) 2024.

TABLE IX
TABLE OF CHANGES IN VEGETATION UTILIZATION TYPES IN WILLOW COUNTY FROM 2020 TO 2024

Class	Area (km^2)			Area Change Rate(%)		
	2020	2022	2024	2020-2022	2022-2024	2020-2024
Buildup	158.59	97.66	89.56	-38.42	-8.29	-43.53
River	57.78	209.35	223.40	262.32	6.71	286.64
Kumquat	129.22	121.97	151.75	-5.61	24.42	17.44
Tree	1334.19	1017.85	1540.99	-23.71	51.40	15.50
Pond	28.67	92.65	19.38	223.16	-79.08	-32.40
Barren	959.39	1042.42	574.55	8.65	-44.88	-40.11
Other	229.84	315.79	296.85	37.40	-6.00	29.16

effectiveness of its dynamic feature adaptation mechanism. The accuracy of the Barren category under MDCA-NET exhibits a steady increase from 97.81% to 98.19% over the three years, in contrast to traditional methods and some DL models. This further underscores the dependence of complex texture classification on high-order feature extraction. With its dynamic feature optimization, MDCA-NET emerges as the preferred methodology for multitemporal feature classification tasks, particularly excelling in complex scenes. The spatial distribution of land use for the three years is shown in Fig. 12.

The changes in vegetation utilization types within Liucheng County from 2020 to 2024 are detailed in Table IX. The data reveal that the area designated for tree coverage experienced fluctuations, decreasing from 1 334.19 km^2 in 2020 to 1 017.85 km^2 in 2022, before rising to 1 540.99 km^2 in 2024. This represents an overall growth rate of 15.50%, with a significant increase of 51.40% occurring between 2022 and 2024, thereby highlighting the effectiveness of afforestation efforts during this latter period. In contrast, the area classified as barren initially increased before subsequently declining, resulting in an overall reduction of 40.11%, with a notable contraction of 44.88% following 2022. This trend suggests a spatial substitution relationship with the rapid expansion of tree coverage, thereby demonstrating the efficacy of ecological restoration initiatives in managing barren land. The area designated for built-up land continues to decrease, while the area allocated for kumquat cultivation exhibits a pattern of decline followed by an increase, culminating in an overall growth of 17.44%. Collectively, these findings indicate that the study area reflects a dual dynamic of ecological restoration alongside the regulation of human activities, contributing to a relatively stable agricultural system.

The dynamics of Kumquat and Tree cultivation from 2020 to 2024 are depicted in Fig. 13. The area allocated for Kumquat cultivation demonstrates a phase-dependent fluctuation, characterized by an initial contraction followed by a

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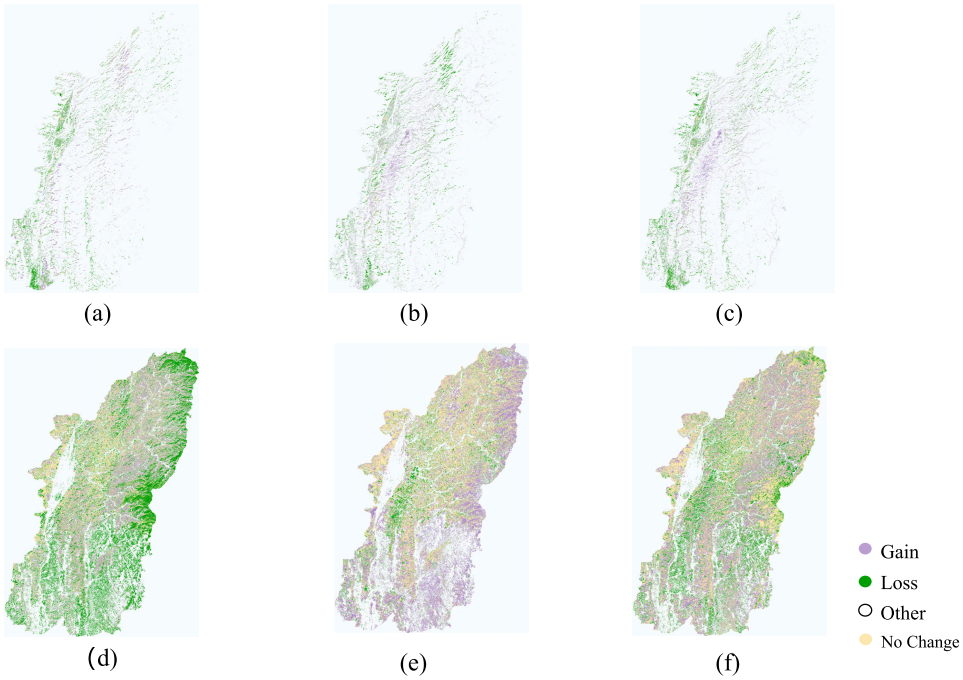


Fig. 13. Dynamics of Kumquat in (a) 2020–2022, (b) 2022–2024, and (c) 2020–2024. Dynamics of forest in (d) 2020–2022, (e) 2022–2024, and (f) 2020–2024.

significant rebound and an overall trend of expansion. The growth observed in the latter period can be attributed to supportive agricultural policies and improved irrigation conditions. Notably, the spatial distribution of Kumquat remains predominantly concentrated in the western and central regions of the study area. In contrast, the coverage of Tree experienced a reversal, initially contracting before undergoing rapid expansion. The early reduction in area may have been influenced by natural disturbances; however, the subsequent rebound indicates the effectiveness of ongoing ecological restoration initiatives. The areas of expansion for Tree significantly overlap with regions of declining bare land, underscoring the successful mitigation of land degradation through vegetation restoration efforts. Spatially, the distribution of Tree extends from its core areas in the East and North toward the South, thereby establishing a continuous ecological corridor that interconnects with the intensive cultivation zones of Kumquat, thereby creating an "ecological–agricultural" interface.

IV. Discussions

This research systematically examines the disparities in classification performance among traditional machine learning algorithms, specifically the SVM and KNN, as well as various DL architectures, including the ViT, CAF, Morphformer, SSFTT, CACFT, and the proposed MDCA-Net. The study utilizes hyperspectral and multispectral remote sensing datasets. The experimental results indicate that MDCA-Net exhibits significant advantages across both types of datasets, which can be attributed to its dynamic feature interaction mechanism and cross-scale modeling capabilities. In contrast, while the SVM and KNN demonstrate high training efficiency and low computational resource requirements, their dependence on manual feature-based modeling restricts their ability to capture complex nonlinear correlations between spectral bands, resulting in notable performance limitations, particularly in hyperspectral contexts. Among the DL methodologies, the ViT enables global spectral correlation modeling through self-attention and positional encoding mechanisms; however, it is burdened by high computational complexity and limited adaptability to small sample sizes. Notably, CAF and Morphformer demonstrate competitive performance, ranking second only to MDCA-Net in multispectral classification tasks, achieved through multiscale feature fusion or optimization of morphological operators, although their generalization capabilities across different datasets remain constrained. SSFTT and CACFT typically rely on static attention mechanisms or fixed-scale feature extraction methods, which restrict their adaptability and limit their capacity to enhance contextual information in complex geographic feature scenarios. The MDCA-Net enhances local detail and models global dependencies within a lightweight architecture by integrating dynamic convolution with a self-attention mechanism. This integration ultimately facilitates the synergistic optimization of local detail enhancement and global dependency modeling in complex geographic regions, thereby yielding superior performance in intricate feature classification and cross-sensor adaptation.

Despite its advantages, the MDCA-Net model presents certain limitations. The pseudo-temporal generation process is dependent on manual replica expansion, which has the potential to introduce extraneous noise. Moreover, the utilization of single-pixel input fails to capture authentic spatial neighborhood information, thereby limiting the model's ability to finely discriminate complex feature boundaries. Additionally, the cascade architecture of BiLSTM networks, augmented by multiple attention mechanisms, improves feature representation but results in increased computational costs, which may impede the real-time processing of high-resolution remote

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sensing images. Future improvements could involve the integration of lightweight spatial interpolation techniques to merge local spatial contexts with minimal computational overhead. Furthermore, the investigation of a joint compression architecture that combines attention mechanisms with LSTM could facilitate a balance between performance and efficiency, thus meeting the deployment requirements of edge computing scenarios.

V. Conclusion

This research delineates the development of a hybrid neural network model, designated as MDCA-NET, which amalgamates multiscale convolution, an attention mechanism, and temporal modeling to enhance effective feature extraction and classification through the synergistic interaction of multiple modules. The foundational architecture of the model is predicated on variants of RNNs, specifically LSTM networks and MLPs. Initially, the input data is processed through the D-MLKA module, which executes multiscale feature fusion with dynamic weight adjustment. This module utilizes various sizes of dilated convolution to capture both local and global information while preserving original features through residual connections. Simultaneously, the SS-CAFM module augments cross-channel and spatial feature interactions by integrating global attention derived from 3-D convolution with local convolutional branching, further refined by spectral–spatial synergistic attention.

Subsequently, the feature-enhanced data undergo pseudo-temporal modeling, during which temporal dependencies are captured through the implementation of BiLSTM networks and an attention mechanism. The classification process is ultimately performed using an MLP. The model exhibited commendable accuracy on publicly available datasets, achieving an overall accuracy of 77.38% on the Indian Pines dataset and 83.34% on the Houston University dataset. Furthermore, the model was employed to classify features in Rong'an County, yielding similarly impressive results, thereby demonstrating the versatility of the MDCA-NET.

It is essential to acknowledge that the MDCA-NET model is distinguished by a considerable number of parameters and significant computational complexity, which may impede its efficiency in real-time deployment contexts. Future research could concentrate on reducing computational overhead by employing lightweight design strategies or integrating efficient attention mechanisms to more effectively meet the demands of edge computing and other low-power applications.

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